

Developments in pipeline for the future

"Altera is working on development kits based on Stratix 10—a new family of 14nm FPGAs built on FinFET technology with integrated ARM A53 hardened quad-cores."

—Natarajan M.M., vice president for South Asia of Arrow Asia-Pacific components business

"Xilinx has developed their latest FPGA families, UltraScale 16nm and 20nm. Avnet is actively working with this new silicon technology, and we are excited to introduce these new products based on this technology in the near future."

—Bryan Fletcher, global technical marketing director, Avnet Electronics Marketing

"Our exclusive manufacturing partnership with Intel opens up avenues for innovations and enables us to provide a greater degree of integration. With multi-die 3D integration of static random access memory (SRAM), dynamic random access memory (DRAM) and ASICs in our generation 10 portfolio, capabilities and options are endless."

—Prasad Reddy, FAE manager, South Asia Pacific, Altera

"Quad Small Form Factor Pluggable+ (QSFP) interface and VPX (an American National Standards Institute standard) would be incorporated in future products."

—Bailey Hsu, director - marketing division, Terasic

The new era of FPGA systems can hold up to 100 million gates that help us to bring a larger application into a handful size of device

mable logic, informs Bryan Fletcher, global technical marketing director, Avnet Electronics Marketing. He says, "With the added sophistication of this device, we recognised that many engineers would appreciate having a Zynq based SoM that could easily be adopted into their systems."

Adding to that, Fletcher says, "Currently, we have four Zynq based SoM product families, which will enable companies to reduce risk and increase their time-to-market."

Another addition features new SoC FPGAs, that is, FPGAs with hard processors built-in. "While these are really cool, these are also fairly complicated to work with and definitely not a recommended starting point for beginners. These are great because FPGAs and processors are generally good at things the other is bad at," says Justin Rajewski, CEO, Embedded Micro. So, having both can be helpful.

He adds, "Beginners can experiment with our Mojo board, where they get to program the Arduino (AT-

mega32U4) and the FPGA (XC6SLX9), as it aims to simplify getting started with FPGA and digital designing."

Talking from Terasic's point of view, Bailey Hsu, director - marketing division, Terasic, notes, "Latest improvements in our FPGA based development boards are covered in two different directions. One is smaller in size but performs faster, which can be seen in DES-NEEK kits scheduled to be available around the third quarter of 2015. Other is cheaper but more versatile displayed in DE0-Nano-SoC boards."

He adds, "We are currently developing platforms with VPX and high-frequency trading (HFT) and will focus more on these two fields in future."

Altera too offers more solution-centric development boards for applications from audio/video, military solutions, 40G/100G applications, to name a few, informs Prasad Reddy, FAE manager, South Asia Pacific, Altera. He says, "With the addition of high-speed mezzanine card (HSMC) connectors, designers can leverage pluggable solution-specific daughter cards from both Altera and our partners to address specific solution needs."

Although this is not a new area, there have been significant additions with respect to the compatibility of expansion boards with FPGA and SoC

based development boards. Natarajan M.M., vice president for South Asia of Arrow Asia-Pacific components business, says, "The new-generation FPGA boards are equipped with HSMC/FPGA mezzanine card (FMC) connectors, allowing add-on kits from third-party boards to be plugged on base boards designed on FPGAs."

On similar lines, Fletcher says, "Two expansion standards that we have seen the most are FMC and Digilent-compatible peripheral module (Pmod). An FMC has many I/Os than Pmod, which means, it has the ability to contain much more circuitry."

He adds, "Pmods are smaller and lower in cost. We have even adapted Pmod standard to pass high-speed differential signals for embedded touch displays."

FPGAs fuelling faster time-to-market, high-priority applications

Adding support for various configurations of external daughter cards, network processing and industrial control modules have enabled users to use a common platform for applications having varied signal acquisition, interfacing and performance targets. With reusability being a much sought-after requirement today, supporting the use of a common FPGA based prototyping environment for different designs has significant cost and time-to-market advantages.

Kumar says, "FPGA with DSP features is another area that is enabling applications requiring comprehensive signal processing."

He adds, "Integration of configurable SoC blocks to FPGA architecture is another area that has resulted in quick turnaround of prototyping complex systems and sub-systems in an FPGA."

Kumar informs, "Though low-power FPGA architectures are enabling handheld applications such as mobile computing and wearable medical electronics, the industry expects better performance and power ratio than available today to really tap into this market." He adds, "High-performance FPGA architecture from Xilinx/Altera,